1 Signal Plotting

```
clc;

% Sine Signal

subplot(3, 2, 1);
t = 0: 0.000001:pi;
y = sin(2*pi*t);
xlabel('Time');
ylabel('Amplitude');
plot(y);

% cosine Signal

subplot(3, 2, 2);
t = 0: 0.000001:pi;
y = cos(2*pi*t);
xlabel('Time');
ylabel('Amplitude');
plot(y);

% Exponential Signal

subplot(3,2,3);
t = -5:0.01:5;
y = exp(t);
plot(t,y);
xlabel('Time');
ylabel('Amplitude');
title('Exponential Signal');

% Impulse Signal

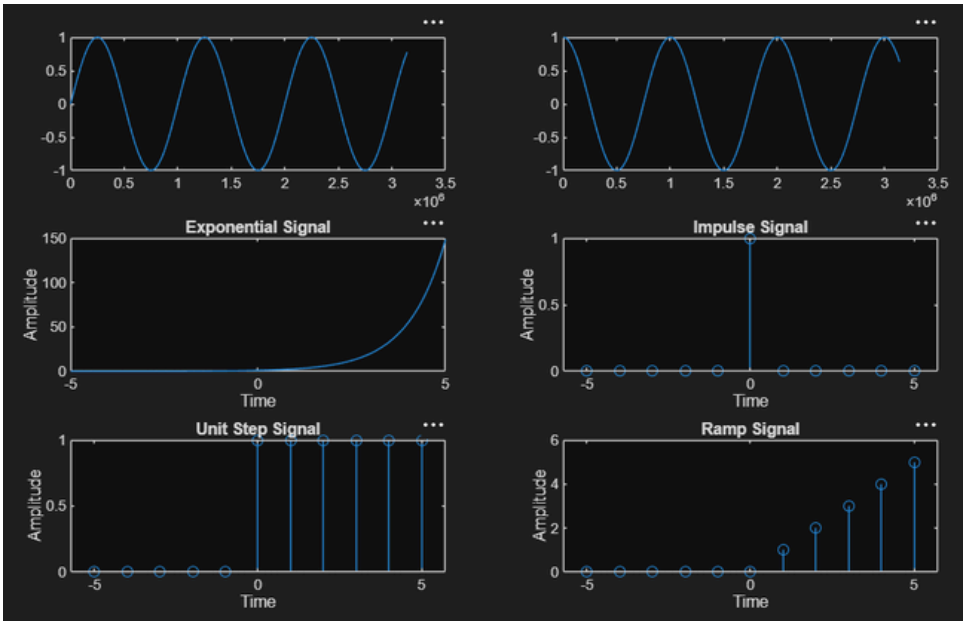
subplot(3,2,4);
t = -5:1:5;
y = zeros(size(t));
y(t==0) = 1;
stem(t,y);
xlabel('Time');
ylabel('Amplitude');
title('Impulse Signal');

% Unit Step Signal

subplot(3,2,5);
t = -5:1:5;
y = (t>=0);
stem(t,y);
xlabel('Time');
ylabel('Amplitude');
title('Unit Step Signal');

%ramp signal

subplot(3, 2, 6);
t = -5:1:5;
y = t.*(t>=0);
stem(t,y);
xlabel('Time');
ylabel('Amplitude');
title('Ramp Signal');
```



2 DFT and IDFT

```
clc;

x=[1,2, 0, 0];
h=[2,2,0,0];
N = length(x);

X=fft(x);
H=fft(h);

Y=H.*X;
y=ifft(Y);

t=0:1:N-1;
clc;
subplot(3,2,1);
stem(t, x, 'filled', 'LineWidth',
1.5);
title('Input Signal');

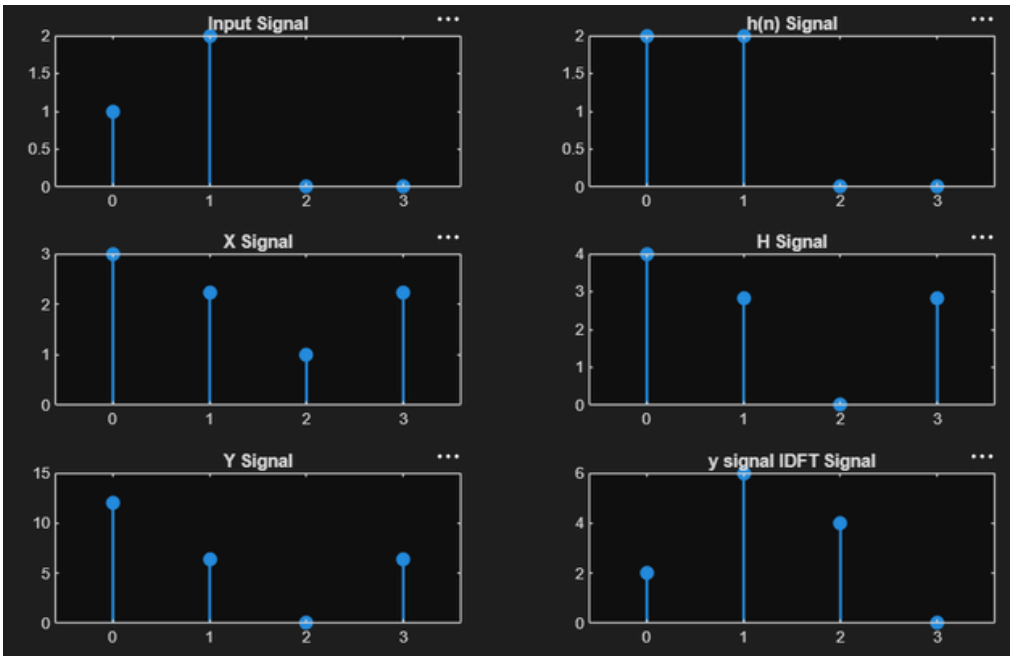
subplot(3,2,2);
stem(t, h, 'filled',
'LineWidth', 1.5);
title('h(n) Signal');

subplot(3,2,3);
stem(t, abs(X), 'filled',
'LineWidth', 1.5);
title('X Signal');

subplot(3,2,4);
stem(t, abs(H), 'filled',
'LineWidth', 1.5);
title('H Signal');

subplot(3,2,5);
stem(t, abs(Y), 'filled',
'LineWidth', 1.5);
title('Y Signal');

subplot(3,2,6);
stem(t, real(y), 'filled',
'LineWidth', 1.5);
title('y signal IDFT Signal');
```



3 Circular Convolution

```
clc;
clear;

x=[1,2,0,1];
h=[2,2,1,1];

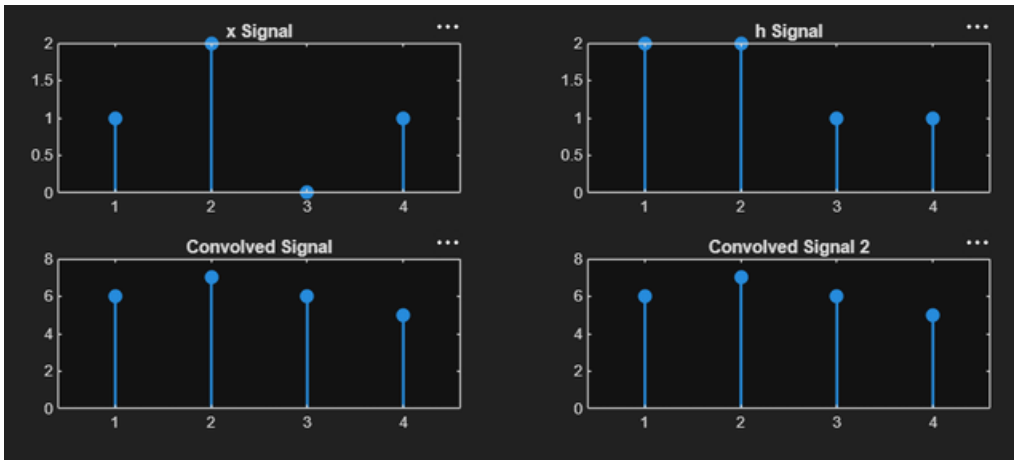
N=length(x);

y1= cconv(x,h,N);

subplot(3,2,1);
stem(x, 'filled', 'LineWidth', 1.5);
title('x Signal');

subplot(3,2,2);
stem(h, 'filled', 'LineWidth', 1.5);
title('h Signal');

subplot(3,2,3);
stem(y1, 'filled', 'LineWidth', 1.5);
title('Convolved Signal');
```




```
%Expansion

X=fft(x);
H=fft(h);
Y=X.*H;

y2 = ifft(Y);

clc;
subplot(3,2,4);
stem(y2, 'filled', 'LineWidth', 1.5);
title('Convolved Signal 2');
```

4 Moving Average Filter

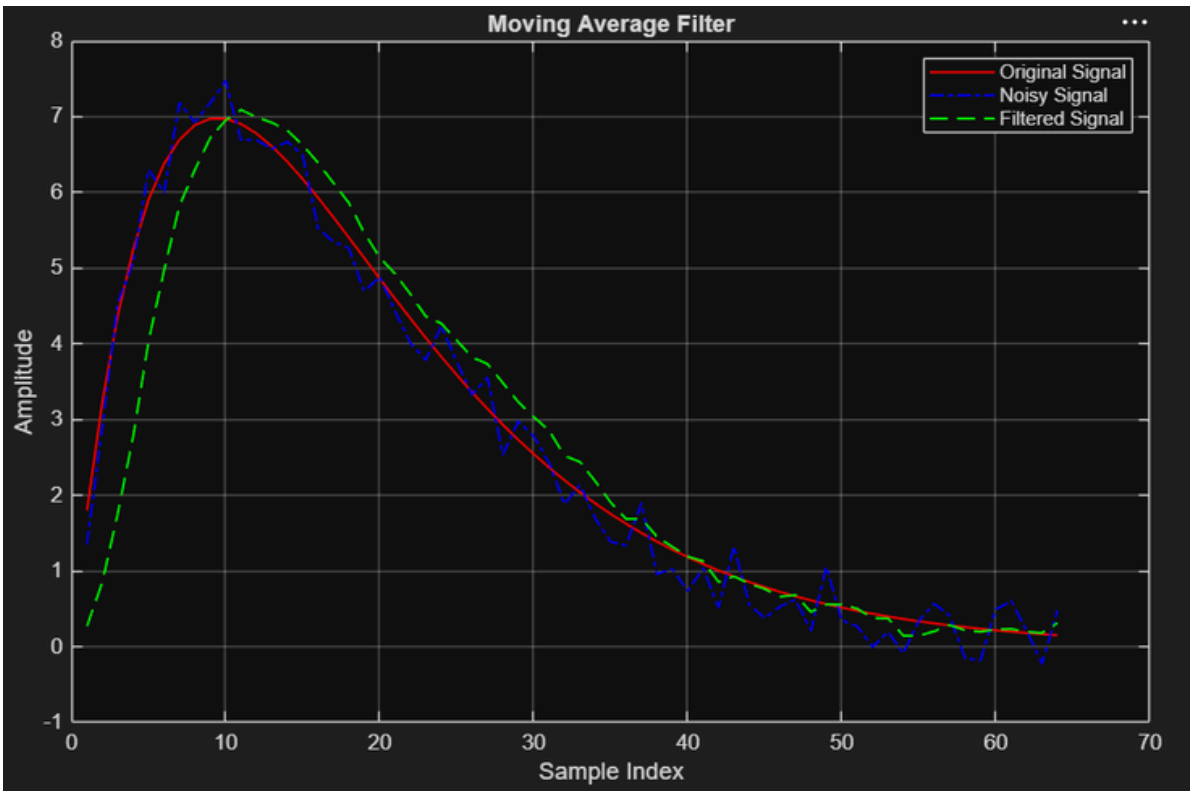
```
clc;
clear;
close all;

r = 64;
d = rand(1,r)-0.5;
k = 1:r;

s = 2*k.*(0.9.^k);
n = length(s);
x = s+d;
M=input('Enter a Number: ');
h=ones(1,M)/M;
y=fftfilt(h,x,n);

plot(s,'r-','LineWidth', 1);
hold on;
plot(x,'b-.','LineWidth', 1);
plot(y,'g--','LineWidth', 1);

legend('Original Signal','Noisy Signal','Filtered Signal');
title('Moving Average Filter');
xlabel('Sample Index');
ylabel('Amplitude');
grid on;
```



5 Median Filter

```
clear all;
close all;
clc;

r = 64;
d = rand(1,r)-0.4;
dr = round(d);

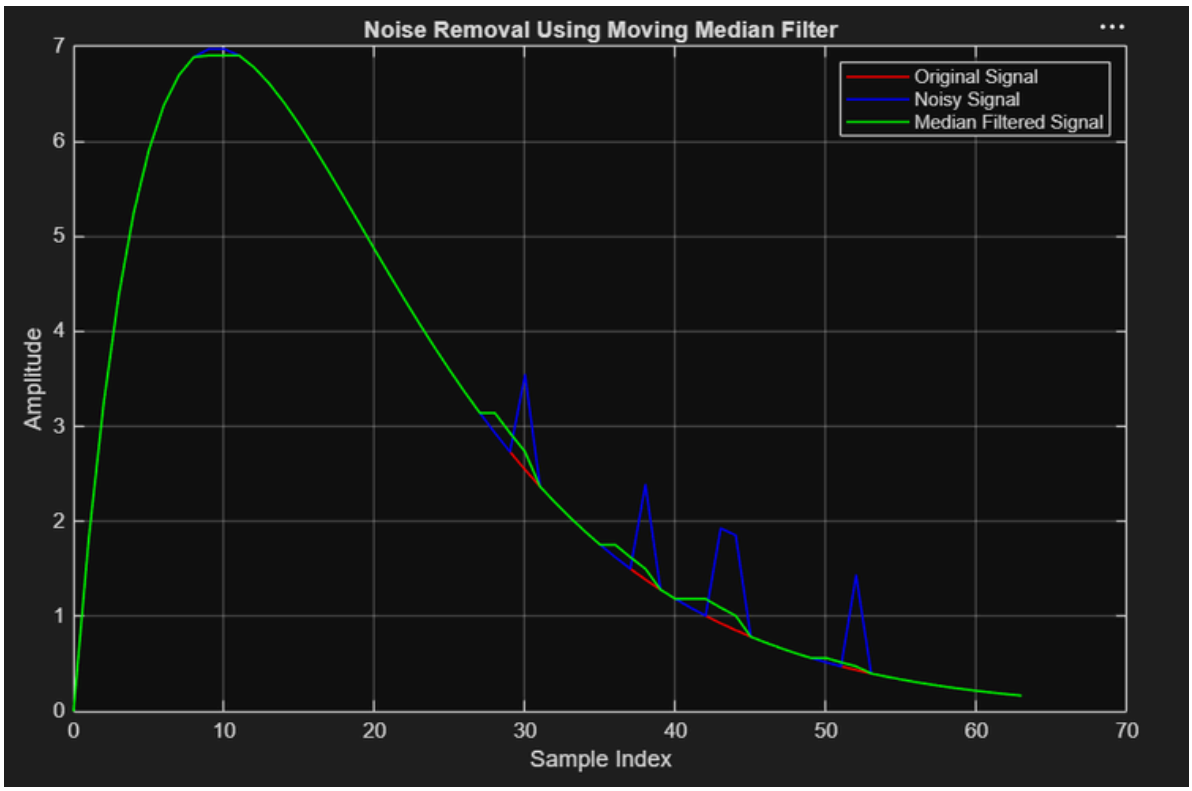
k = 0:1:r-1;
s = 2*k.*(0.9.^k);
x = s + dr;

M = input('Enter an odd window size (e.g., 3 or 5): ');

y = medfilt1(x, M);

plot(k, s, 'r-', 'LineWidth', 1);
hold on;
plot(k, x, 'b-', 'LineWidth', 1);
plot(k, y, 'g-', 'LineWidth', 1);

legend('Original Signal', 'Noisy Signal', 'Median Filtered Signal');
title('Noise Removal Using Moving Median Filter');
xlabel('Sample Index');
ylabel('Amplitude');
```



6 Low pass and High pass filter

```
clc;
clear all;
close all;

b = [6.762, -13.524, 6.762];
zi = [0 0];
n = 0:99;
x1 = cos(0.1*n);
x2 = cos(0.4*n);
y = filter(b, 1, x1+x2, zi);

plot(n, x1, 'r');
hold on;
plot(n, x2, 'b');
plot(n, x1+x2, 'm');
plot(n, y, 'g');

legend('x1 - Low Frequency', ...
 'x2 - High Frequency', ...
 'Combined Input', ...
 'Filtered Output');

xlabel('Sample Index (n)');
ylabel('Amplitude');
title('FIR Low-Pass Filter');
grid on;
```

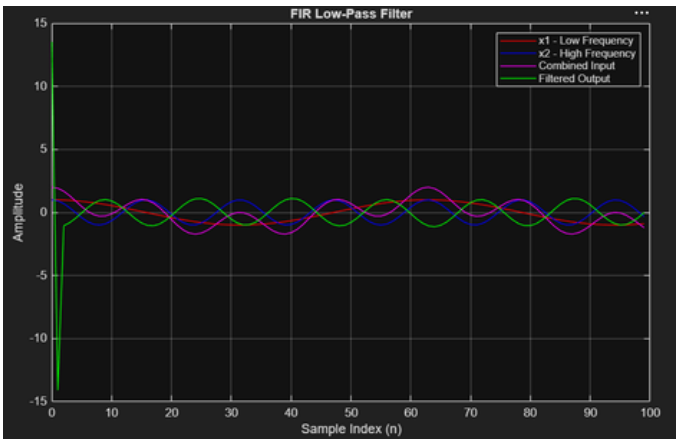
```
clc;
clear all;
close all;

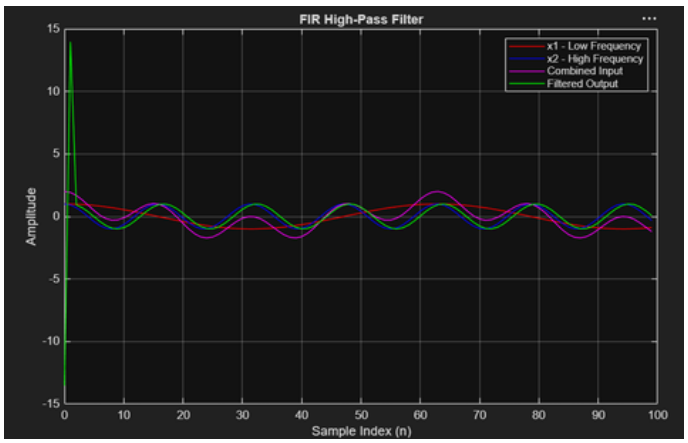
b = [-6.762, 13.456, -6.762];
zi = [0 0];
n = 0:99;
x1 = cos(0.1*n);
x2 = cos(0.4*n);
y = filter(b, 1, x1+x2, zi);

plot(n, x1, 'r');
hold on;
plot(n, x2, 'b');
plot (n, x1+x2, 'm');
plot(n, y, 'g');

legend('x1 - Low Frequency', ...
 'x2 - High Frequency', ...
 'Combined Input', ...
 'Filtered Output');

xlabel('Sample Index (n)');
ylabel('Amplitude');
title('FIR High-Pass Filter');
grid on;
```





7 Gibbs Low pass filter

```
% Gibbs Phenomenon - Low
Pass Filter
clear all;
close all;
clc;

wc = 0.5*pi;
N = 95;
alpha = (N-1)/2;
eps = 0.001;

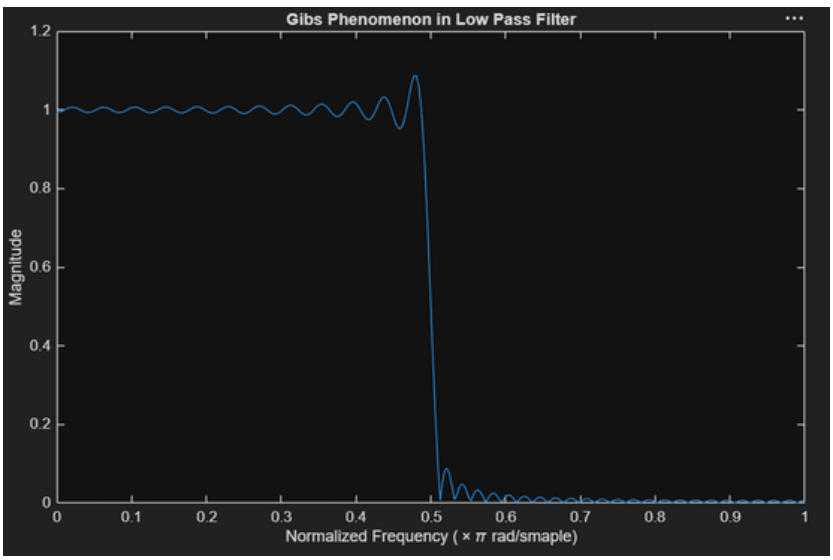
n = 0:1:N-1;

%Ideal low pass filter
hd1 = sin(wc*(n-alpha+eps));
hd2 = pi*(n-alpha+eps);
hd = (hd1)./(hd2);

%Filter coefficient (Rectangular
Window)
hn = hd;

% Frequency response
w = 0:0.01:pi;
h = freqz(hn,1,w);

plot(w/pi, abs(h));
xlabel('Normalized Frequency
(\times\pi rad/smaple)');
ylabel('Magnitude');
title('Gibs Phenomenon in Low
Pass Filter');
```



8 Gibbs High pass filter

```
% Gibbs Phenomenon - High
Pass Filter
clear all;
close all;
clc;

wc = 0.5*pi;
N = 95;
alpha = (N-1)/2;
eps = 0.001;

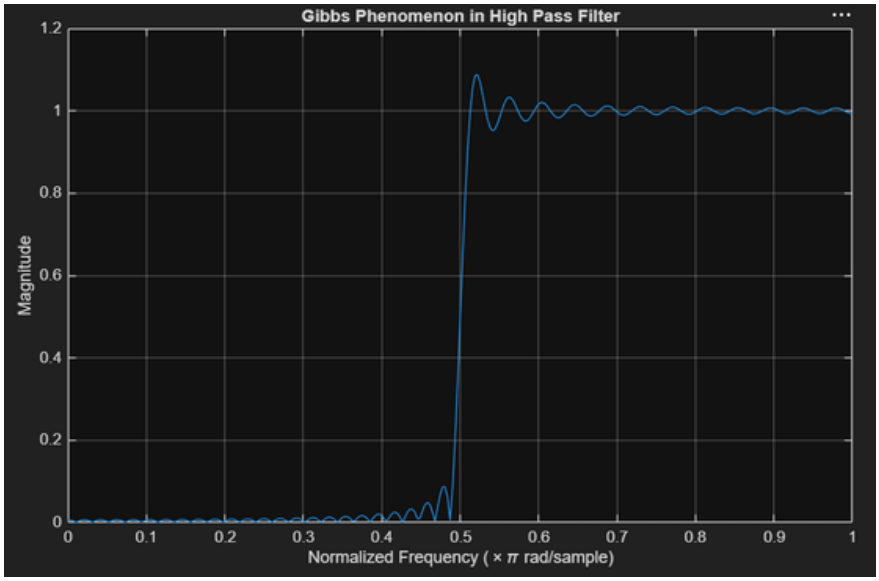
n = 0:1:N-1;

%Ideal high pass filter
hd1 = sin(pi*(n-alpha+eps)) -
sin(wc*(n-alpha+eps));
hd2 = pi*(n-alpha+eps);
hd = hd1./hd2;

%Filter coefficient
hn = hd;

% Frequency response
w = 0:0.01:pi;
h = freqz(hn,1,w);

plot(w/pi, abs(h));
xlabel('Normalized Frequency
(\times\pi rad/sample)');
ylabel('Magnitude');
title('Gibbs Phenomenon in High
Pass Filter');
grid on;
```



9 Windowing Technique

```
clc;
wc = 0.5*pi;
N = 55;
alpha = (N-1)/2;
eps = 0.001;
n = 0:1:N-1;

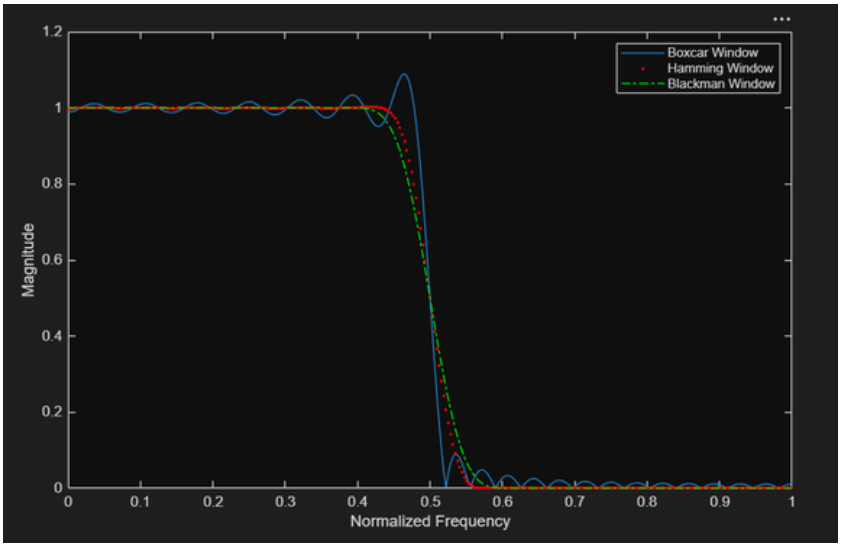
hd1 = sin(wc * (n - alpha + eps));
hd2 = pi*(n - alpha + eps);
hd = (hd1)./(hd2);

wr = boxcar(N);
hn = hd.*wr';
w = 0:0.01:pi;
h = freqz(hn, 1, w);
plot(w/pi, abs(h));
hold on;

wh = hamming(N);
hn = hd.*wh';
w = 0:0.01:pi;
h = freqz(hn,1,w);
plot(w/pi, abs(h), 'r.');
hold on;

wb = blackman(N);
hn = hd.*wb';
w = 0:0.01:pi;
h = freqz(hn,1,w);
plot(w/pi, abs(h), 'g-.');
hold on;

legend('Boxcar Window',
'Hamming Window', 'Blackman
Window');
xlabel('Normalized Frequency');
ylabel('Magnitude')
hold on;
```



10 Kaiser Window

```
clc;

% Given specifications
ap = 0.1;
as = 44;
ws = 30;
wp = 20;
wfs = 100;

% Calculations
B = ws - wp;
wc = 0.5*(wp + ws);
wcr = wc*2*pi/wfs;

D = (as - 7.95)/14.36;
N = ceil((wfs*D/B) + 1);

alpha = (N-1)/2;

gamma = 0.584*(as-21)^0.4 +
0.07886*(as-21);

% Ideal low-pass filter
n = 0:1:N-1;

a1 = sin(wcr*(n-alpha));
a2 = pi*(n-alpha);

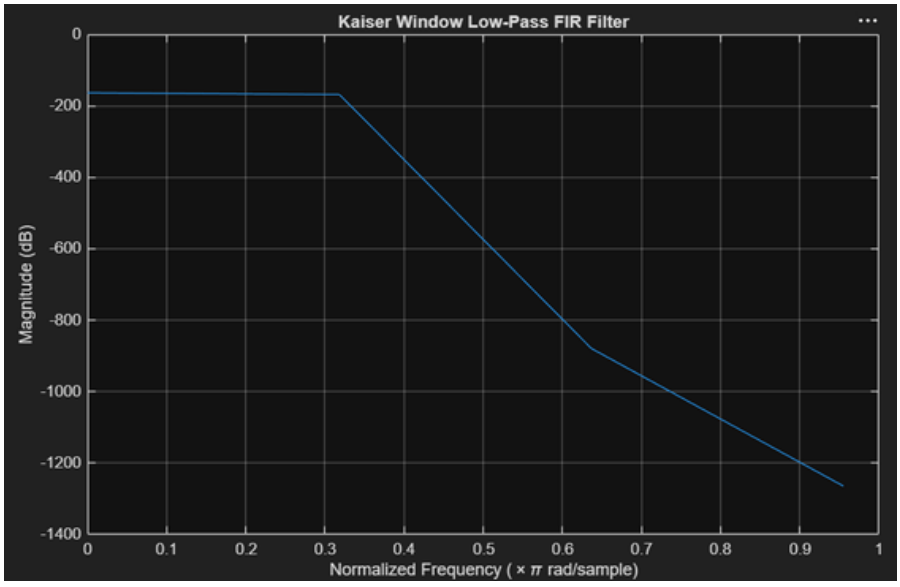
hd = a1./a2;
hd(alpha+1) = 0.5;

% Kaiser window
wk = kaiser(N,gamma);

% Windowed filter coefficients
hn = hd.*wk;

% Frequency response
w = 0:1:pi;
h = freqz(hn,1,w);

% Plot magnitude response in dB
plot(w/pi,20*log10(abs(h)));
xlabel('Normalized Frequency
(\times\pi rad/sample)');
ylabel('Magnitude (dB)');
title('Kaiser Window Low-Pass FIR
Filter');
grid on;
```



11 UP sampling

```
% Illustration of Upsampling
clear;
close all;
clc;

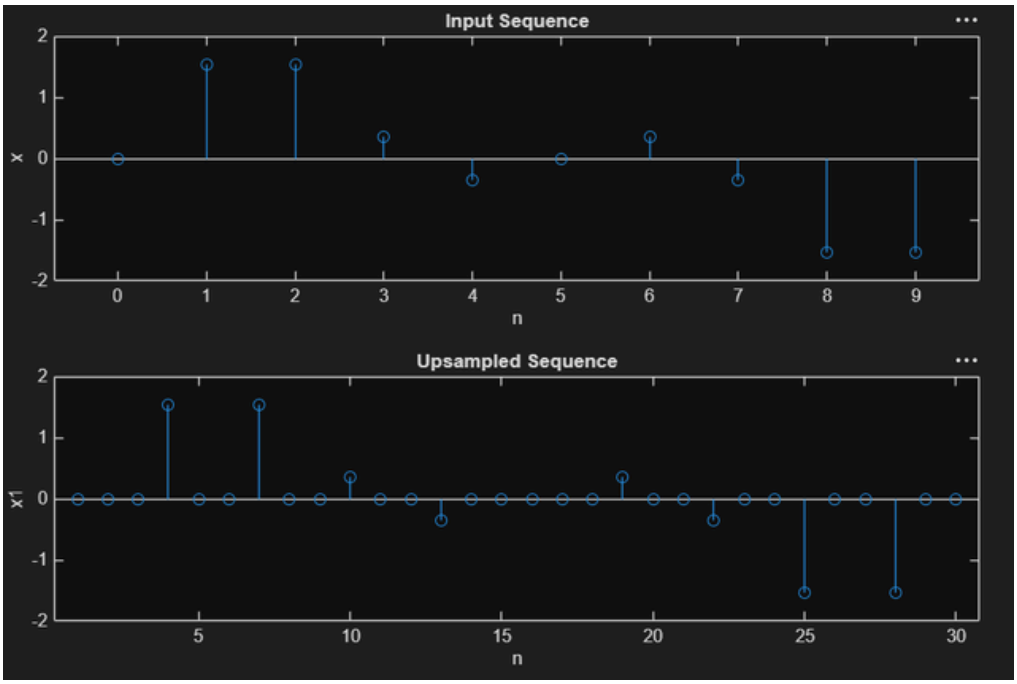
N = 10;
n = 0:1:N-1;

x = sin(2*pi*n/10) + sin(2*pi*n/5);
L = 3; % Upsampling factor

x1 = zeros(1,L*N);
n1 = 1:1:L*N;
j = 1:L:L*N;
x1(j) = x;

subplot(2,1,1);
stem(n,x);
xlabel('n');
ylabel('x');
title('Input Sequence');

subplot(2,1,2);
stem(n1,x1);
xlabel('n');
ylabel('x1');
title('Upsampled Sequence');
```



12 Down sampling

```
% Illustration of Downsampling
clc;

N = 30; % Length of input
sequence
n = 0:1:N-1;

x = sin(2*pi*n/20) +
sin(2*pi*n/15);
M = 3; % Downsampling factor

x1 = x(1:M:N); % Keep every M-th
sample
n1 = 1:1:N/M;

subplot(2,1,1);
stem(n,x);
xlabel('n');
ylabel('x');
title('Input Sequence');

subplot(2,1,2);
stem(n1-1,x1);
xlabel('n');
ylabel('x1');
title('Downsampled Sequence');
```

